

A SIMPLE CONDITIONAL EXCEEDANCE FRAMEWORK FOR INTERPRETABLE TRADING DECISIONS

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ABSTRACT. This paper presents a probability-driven trading framework based on conditional exceedance statistics rather than price or return forecasting. Using daily SPY data, conditional probabilities of large future price moves given large past moves are estimated over fixed horizons using data up to the end of 2024. These probabilities are organized into transparent lookup tables and translated directly into systematic, event-driven buy–hold–sell trading decisions with fixed holding periods. All parameters are frozen prior to evaluation, and performance is assessed strictly out-of-sample on 2025 data. Across multiple parameter configurations, the resulting strategies exhibit strong risk-adjusted performance, with Sharpe ratios exceeding 4 for selected regimes. The results suggest that long-horizon conditional structure in price dynamics can be exploited using simple, interpretable probability tables without reliance on complex predictive models.

1. Introduction

Much of quantitative trading research focuses on prediction: forecasting next-period returns, estimating expected returns, or learning price dynamics using increasingly complex statistical and machine learning models. While such approaches can be effective, they often obscure a simpler and more decision-relevant question: “Given that the market has already experienced a large move over a certain time horizon, how likely is another large move over the same horizon?”

This work is motivated by that question. Rather than attempting to predict prices or returns directly, I frame trading as a conditional probability problem—estimating the likelihood of future price exceedances given past exceedances. The objective is not to maximize in-sample fit or construct a high-frequency signal, but to develop a statistically interpretable framework that maps naturally to trading decisions under uncertainty. The emphasis is on transparency, uncertainty quantification, and strict out-of-sample evaluation[1].

2. Data and Experimental Design

The analysis uses daily adjusted close prices of SPY, a highly liquid equity ETF.

- Training period: inception through December 31, 2024
- Evaluation period: January–December 2025 (out-of-sample)
- Frequency: daily
- Transaction costs: not included (discussed later)

All probability estimates are computed exclusively using pre-2025 data and are frozen prior to evaluation. No parameters are tuned using 2025 data, ensuring a strict separation between learning and testing.

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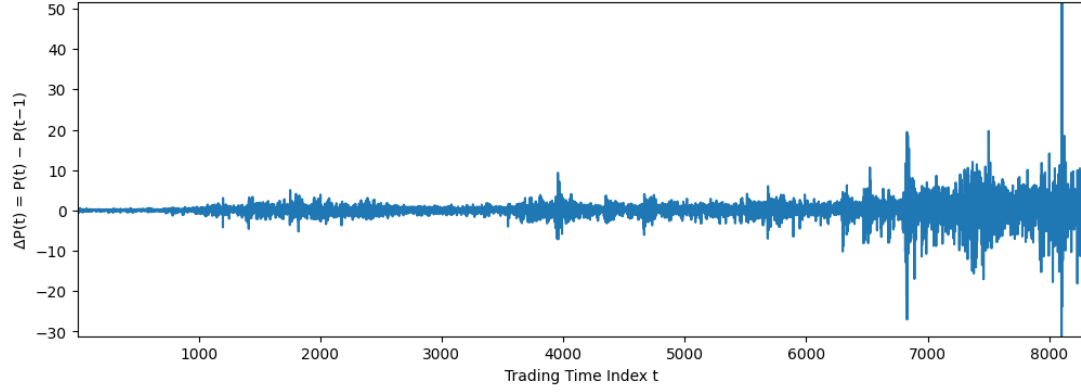


FIGURE 1. Caption

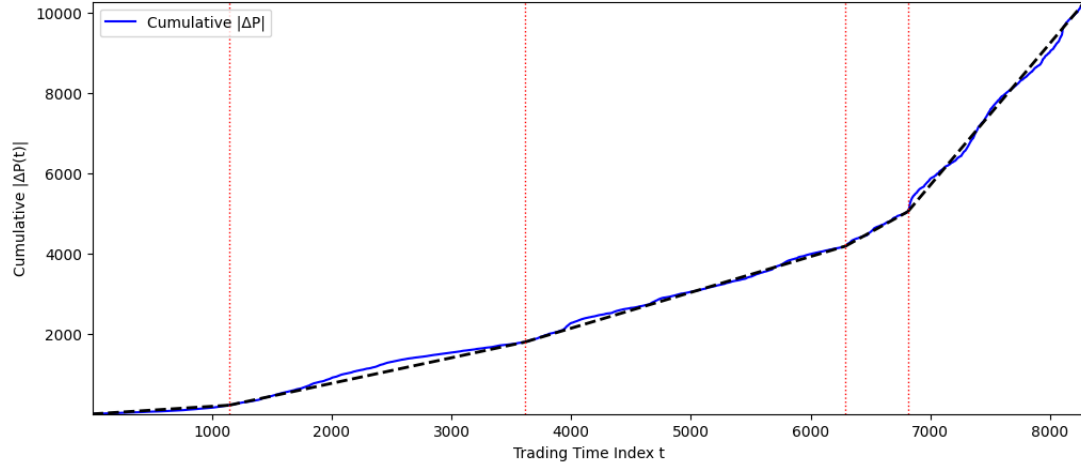


FIGURE 2. Caption

3. Conditional Exceedance Framework

For a fixed time horizon τ , define the τ -day price increment as

$$\Delta P(t; \tau) = P(t) - P(t - \tau)$$

where $P(t)$ denotes the adjusted close price at time t . Rather than modeling returns directly, the analysis focuses on exceedance events:

- A past exceedance occurs when $\Delta P(t; \tau)$ exceeds a high percentile threshold q
- A future exceedance occurs when $\Delta P(t + \tau; \tau)$ exceeds a high percentile threshold r

The central quantity of interest is the conditional probability:

$$Pr[\Delta P(t + \tau; \tau) \geq r | \Delta P(t; \tau) \geq q]$$

By sweeping across values of τ, q, r a multi-dimensional table of conditional exceedance probabilities is constructed. This formulation has several advantages:

- No point forecasting of prices or returns

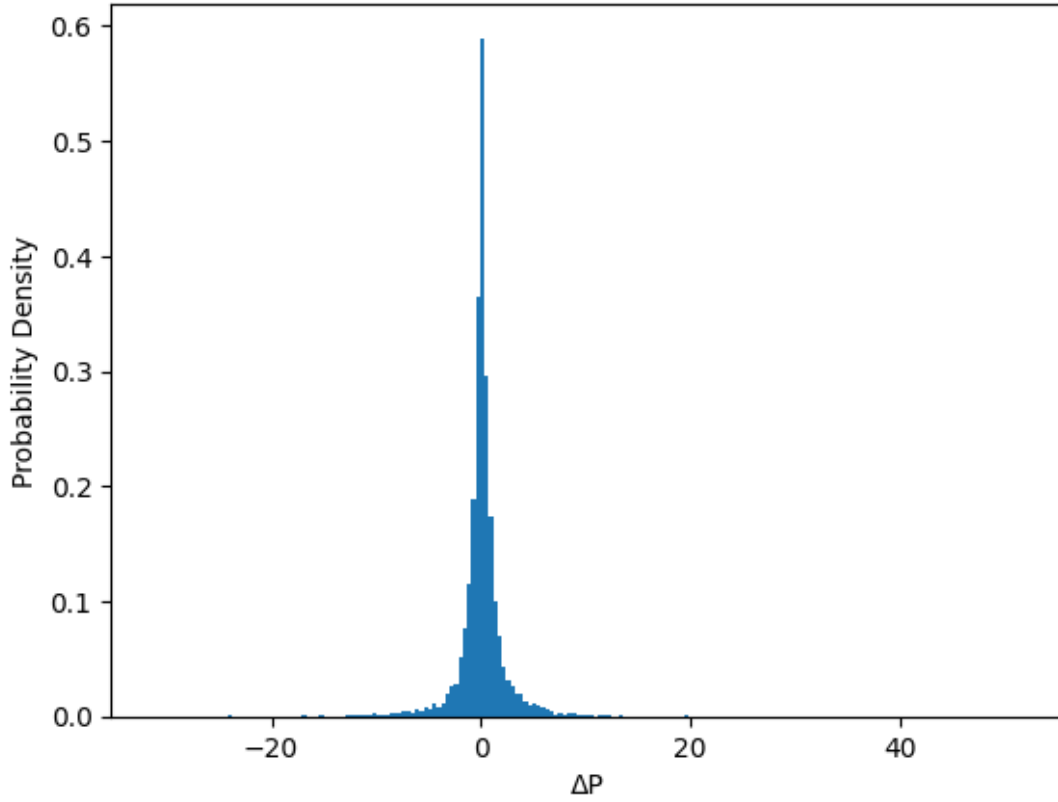


FIGURE 3. Caption

- Direct focus on tail behavior
- A natural mapping from probabilities to trading decisions

4. Statistical Reliability and Uncertainty Control

Conditional probabilities estimated from finite samples can be noisy, particularly in the tails. For each parameter combination N_c : number of conditioning exceedance events is tracked. And each estimate u is treated as a metric of reliability:

$$u = \frac{1}{N_c}$$

Only conditional probabilities satisfying:

- a minimum probability threshold, and
- a maximum uncertainty bound

are permitted to generate trades. This explicitly filters out rare conditioning events, unstable tail estimates, and spurious high-probability artifacts.

5. Trading Strategy Design

The trading rules are intentionally simple and mechanical. On each trading day t in 2025, for a given horizon τ

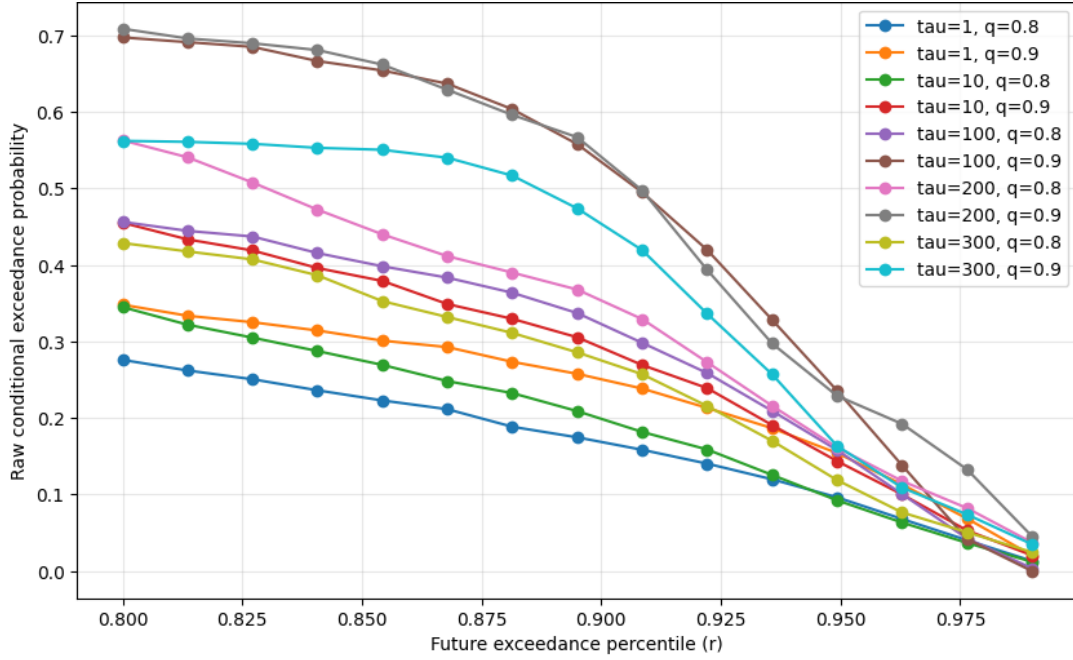


FIGURE 4. Caption

i) heck whether the past τ -day price increment exceeds the threshold q ii) If true, and if the corresponding conditional probability passes statistical filters:

- Enter a long position at time t
- Hold the position for exactly τ trading days
- Exit mechanically at $t + \tau$

Key characteristics of the strategy include:

- No signal aggregation
- No stop-loss or take-profit rules
- Fixed holding horizon per trade
- Overlapping trades allowed
- No parameter tuning on 2025 data

Each trade corresponds to a single, explicit probabilistic hypothesis.

6. Out-of-Sample Results

Performance is evaluated exclusively on 2025 data. Across multiple parameter configurations, the strategy exhibits:

- Sharpe ratios: approximately 4.5–6.5
- Cumulative returns: approximately 650–900 percentage
- Trade counts: several independent trades per configuration

The most robust performance occurs at longer horizons ($\tau = 100$ –300 days). Shorter horizons display weaker and less stable conditional structure, consistent with noise-dominated dynamics at higher frequencies.

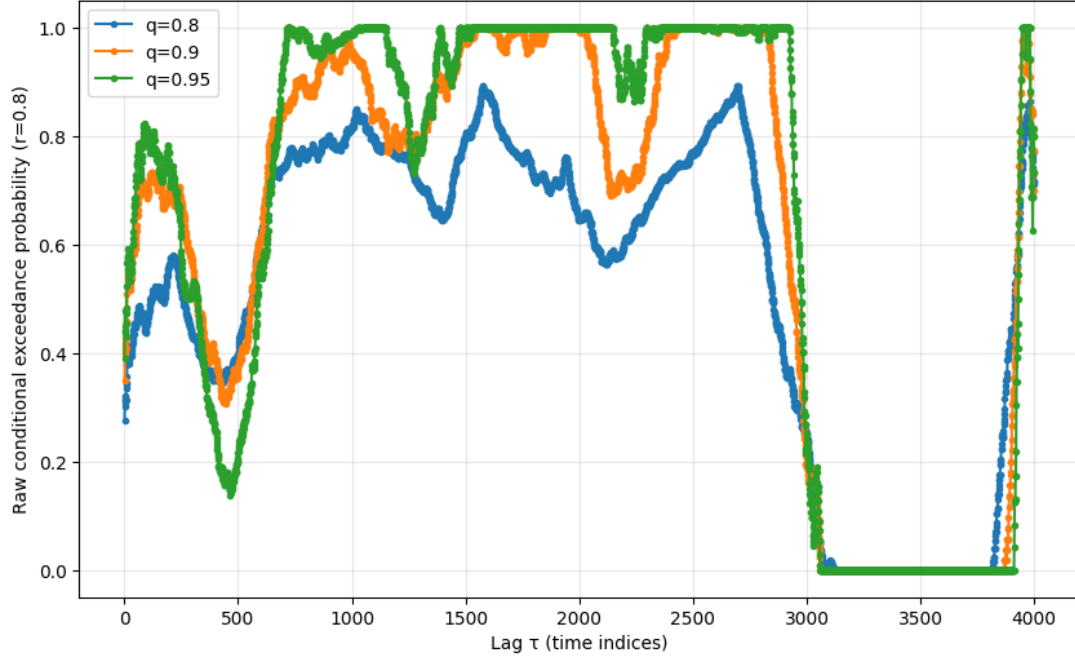


FIGURE 5. Caption

TABLE 1. Out-of-sample performance for selected parameter configurations (2025).

q	r	τ	$p_{\min}$	$u_{\max}$	Trades	Total Return (%)	Sharpe
1	0.8	200	0.5	0.01	55	902.59	4.62
4	0.9	200	0.5	0.01	44	653.16	6.32
3	0.9	100	0.5	0.01	51	245.72	1.89
0	0.8	100	0.5	0.01	0	0.00	—
2	0.8	300	0.5	0.01	0	0.00	—
5	0.9	300	0.5	0.01	0	0.00	—

7. Interpretation

Several points are worth emphasizing:

- The strategy does not predict prices or returns
- It operates entirely on event probabilities
- Every trade is backed by an explicit statistical statement

High risk-adjusted performance arises from consistency in conditional behavior rather than trade frequency or leverage. Because the framework is table-based and transparent, failure modes, confidence levels, and robustness can be analyzed directly through event counts and uncertainty measures.

At a conceptual level, the framework resembles survival or hazard-style modeling applied to financial time series, with exceedances playing the role of events.

8. Limitations

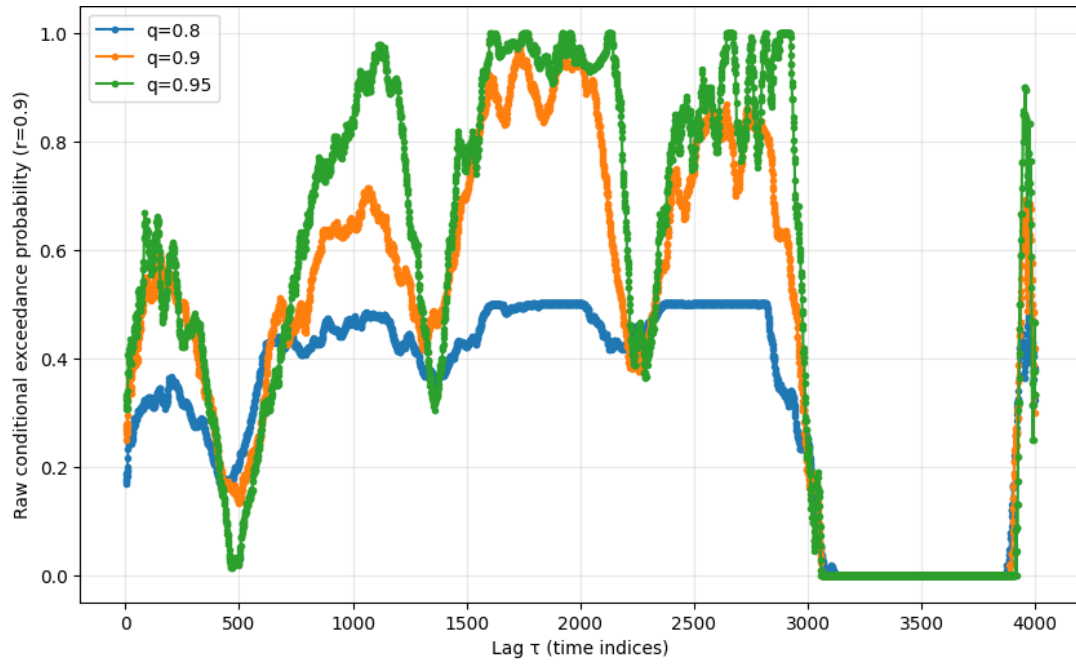


FIGURE 6. Caption

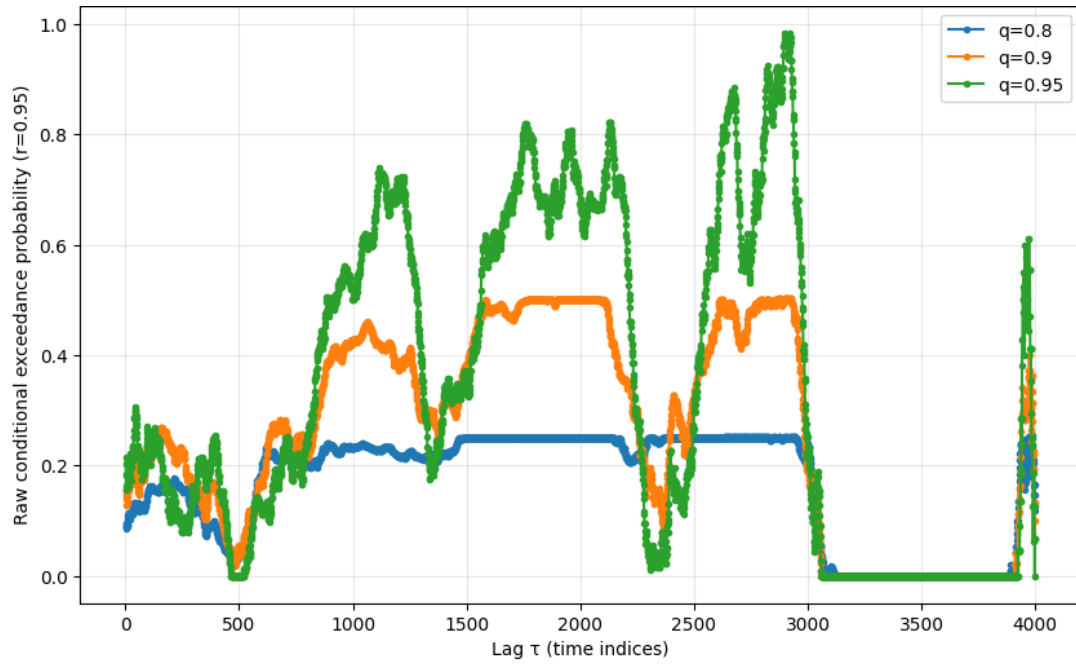


FIGURE 7. Caption

This study is intended as a research prototype and omits several real-world considerations:

- Transaction costs and market impact
- Capital allocation constraints
- Dynamic position sizing
- Portfolio-level interactions

The reported results should therefore be interpreted as evidence of statistical structure rather than deployable production performance.

9. Reproducibility

All probability estimation and trading logic are implemented in Python. A complete code repository, including data preprocessing, probability table construction, and backtesting scripts, is publicly available on GitHub (<https://github.com/quantarram/quant-regime-research>).

10. Conclusion

This paper demonstrates that conditional exceedance probabilities, when estimated and filtered rigorously, can form the basis of a systematic trading framework with strong out-of-sample, risk-adjusted performance.

Rather than asking “what will the price do next?”, the approach asks:

“Given what just happened, how likely is a meaningful move?”

For decision-making under uncertainty, that distinction matters.

References

- [1] Spiegel, M. R. (1981). Theory and problems of Advanced Calculus: Si (metric) edition. McGraw-Hill.

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